CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 6: Solving Problems Based on Quadratic Equations

Topic-wise short notes with solved examples

Many word problems lead to a quadratic equation. This chapter is about turning the words into an equation, solving it, and keeping only the answer that makes sense.

1 The method

- **Step 1**: read the problem and decide what is to be found. Let that quantity be x.
- **Step 2**: write the other quantities in terms of x.
- **Step 3**: use the condition given in the problem to form an equation, and simplify it to $ax^2 + bx + c = 0$.
- **Step 4**: solve the equation.
- **Step 5**: check each root against the problem. Reject a root that is negative when the quantity must be positive, or a fraction when it must be a whole number.
- **Step 6**: write the answer in words, with units.

2 Problems on numbers

- Two consecutive numbers: x and $x + 1$. Two consecutive even (or odd) numbers: x and $x + 2$.
- A number and its reciprocal: x and $\frac{1}{x}$.
- A two-digit number with tens digit x and units digit y is $10x + y$.

Example 2.1 The product of two consecutive natural numbers is 132. Find the numbers.

SOLUTION

Let the numbers be x and $x + 1$. Then $x(x + 1) = 132$, so $x^2 + x - 132 = 0$.

$(x + 12)(x - 11) = 0$, so $x = 11$ (a natural number cannot be -12).

The numbers are 11 and 12.

Example 2.2 The sum of a number and its reciprocal is $\frac{10}{3}$. Find the number.

SOLUTION

$x + \frac{1}{x} = \frac{10}{3}$. Multiply by $3x$: $3x^2 + 3 = 10x$, so $3x^2 - 10x + 3 = 0$.

$(3x - 1)(x - 3) = 0$

The number is 3 or $\frac{1}{3}$.

3 Problems on ages

- If the present age is x years, the age n years ago was $x - n$ and the age n years from now will be $x + n$.

Example 3.1 The product of Ravi's age 3 years ago and his age 5 years from now is 84. Find his present age.

SOLUTION

Let his present age be x years. Then $(x - 3)(x + 5) = 84$, so $x^2 + 2x - 99 = 0$.

$(x + 11)(x - 9) = 0$, so $x = 9$ (age cannot be negative).

Ravi is 9 years old.

4 Problems on geometry

- Rectangle: area $=$ length $\times$ breadth; perimeter $= 2(\text{length} + \text{breadth})$.
- Right-angled triangle: $(\text{hypotenuse})^2 =$ sum of the squares of the other two sides.

Example 4.1 The hypotenuse of a right-angled triangle is 13 cm and the other two sides differ by 7 cm. Find the two sides.

SOLUTION

Let the shorter side be x cm; the other is $(x + 7)$ cm. Then $x^2 + (x + 7)^2 = 13^2$.

$2x^2 + 14x + 49 = 169$, so $x^2 + 7x - 60 = 0$ and $(x + 12)(x - 5) = 0$.

$x = 5$ (a length cannot be negative).

The sides are 5 cm and 12 cm.

Example 4.2 The perimeter of a rectangular field is 46 m and its area is 120 m². Find its dimensions.

SOLUTION

Length + breadth $= 23$. Let the length be x m; the breadth is $(23 - x)$ m.

$x(23 - x) = 120$, so $x^2 - 23x + 120 = 0$ and $(x - 15)(x - 8) = 0$.

The field is 15 m long and 8 m wide.

5 Problems on speed, distance and time

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

- Write the time for each case and use the given difference in time.
- Boat in a stream: speed downstream $= x + y$ and speed upstream $= x - y$, where x is the speed in still water and y the speed of the stream.
- Keep the units consistent: 1 hour 40 minutes $= \frac{5}{3}$ hours.

Example 5.1 A car covers 400 km at a uniform speed. If its speed had been 12 km/h more, it would have taken 1 hour 40 minutes less. Find the speed of the car.

SOLUTION

Let the speed be x km/h. Then $\frac{400}{x} - \frac{400}{x+12} = \frac{5}{3}$.

$\frac{400 \times 12}{x(x+12)} = \frac{5}{3}$, so $x(x+12) = 2880$ and $x^2 + 12x - 2880 = 0$.

$(x+60)(x-48) = 0$, so $x = 48$.

The speed of the car is 48 km/h.

6 Problems on cost and on work

- Cost of one article $= \frac{\text{total cost}}{\text{number of articles}}$.
- If a person finishes a job in x days, the work done in one day is $\frac{1}{x}$.

Example 6.1 A shopkeeper buys a number of books for ₹600. If each book had cost ₹5 less, he could have bought 4 more books for the same amount. Find the number of books he bought.

SOLUTION

Let the number of books be x. Then $\frac{600}{x} - \frac{600}{x+4} = 5$.

$\frac{600 \times 4}{x(x+4)} = 5$, so $x(x+4) = 480$ and $x^2 + 4x - 480 = 0$.

$(x+24)(x-20) = 0$, so $x = 20$.

He bought 20 books.

Example 6.2 A takes 6 days less than B to finish a piece of work. Working together they finish it in 4 days. Find the time B takes alone.

SOLUTION

Let B take x days; then A takes $(x-6)$ days. $\frac{1}{x} + \frac{1}{x-6} = \frac{1}{4}$.

$4(2x-6) = x(x-6)$, so $x^2 - 14x + 24 = 0$ and $(x-12)(x-2) = 0$.

$x = 2$ would make A's time negative, so $x = 12$.

B takes 12 days (and A takes 6 days).

7 Common mistakes

- Not rejecting the root that does not fit the problem.
- Writing the smaller time first, which gives a negative difference. The slower journey takes the longer time.
- Mixing minutes with hours.
- Stopping at the value of x when the question asks for something else, such as both numbers or both dimensions.

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